

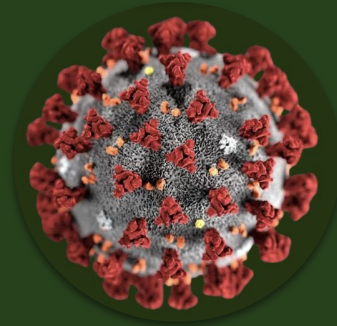
THE SOIL IS SICK.



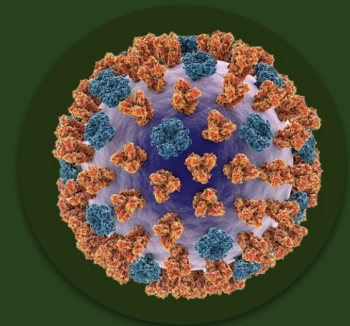
A Computational Investigation of RNA Phages in Soil

RNA bacteriophages & Viruses

- Biological Components
- Ecology
- Viral Importance
- RdRP (RNA-dependent RNA polymerase)



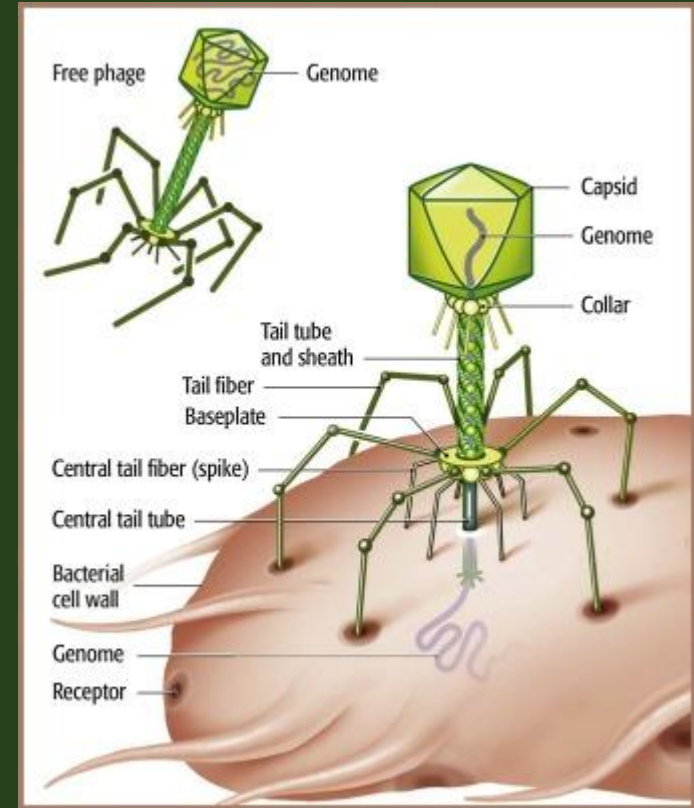
SARS-CoV2 - (UNC)



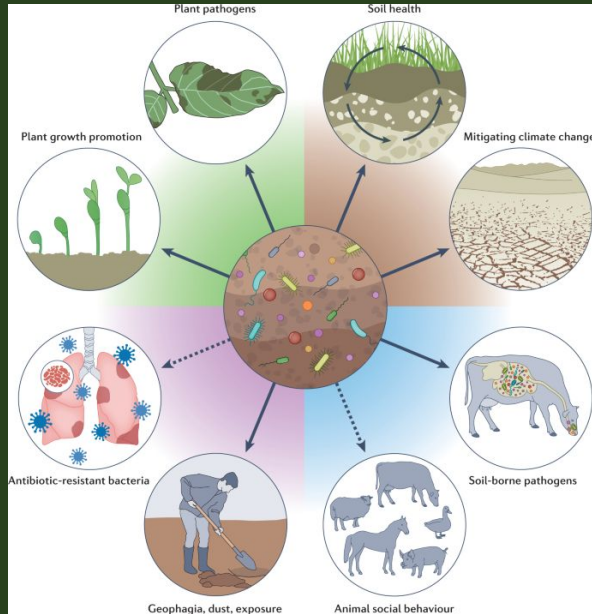
Influenza (CDC)

What are RNA Phages?

- Viruses that infect bacteria
- Most known have single-stranded RNA genomes
- Simple structure:
 - Protein shell (capsid) to protect the RNA
 - RNA-dependent RNA polymerase (RdRP)
 - Host cells do not have an RdRP naturally
→ if found in data, RNA virus was likely there!
 - Coat, maturation, sometimes lysis protein
- Found in:
 - Soil!
 - Human gut, freshwater, the ocean.



Why study soil RNA Phages?



RNA viruses are underrepresented
in viral environmental studies
(mainly DNA focused-
bacteriophages)

- Diverse and abundant,
contributing to microbial
diversity

Mutate faster than bacteriophages,
responding to environmental stress

- Key to understanding how
factors like warming & drought
alter microbial systems

Where Is Our Data from?

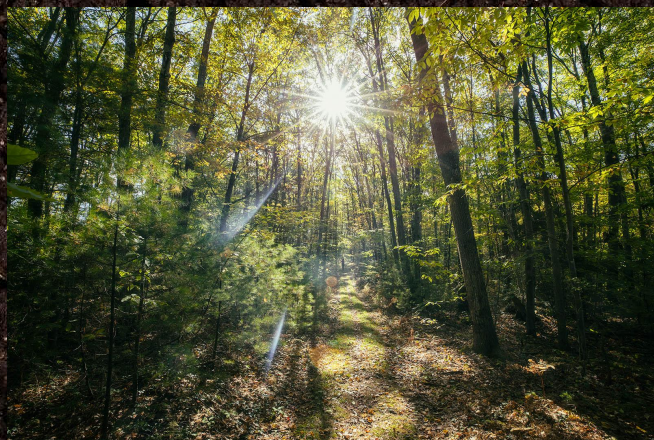
- Soil RNA data from Harvard Forest (via IMG/M)
- Heated (+5°C) vs control plots
- Organic and mineral horizons sampled

Environmental Context

- Temperate forest biome, long-term warming experiment

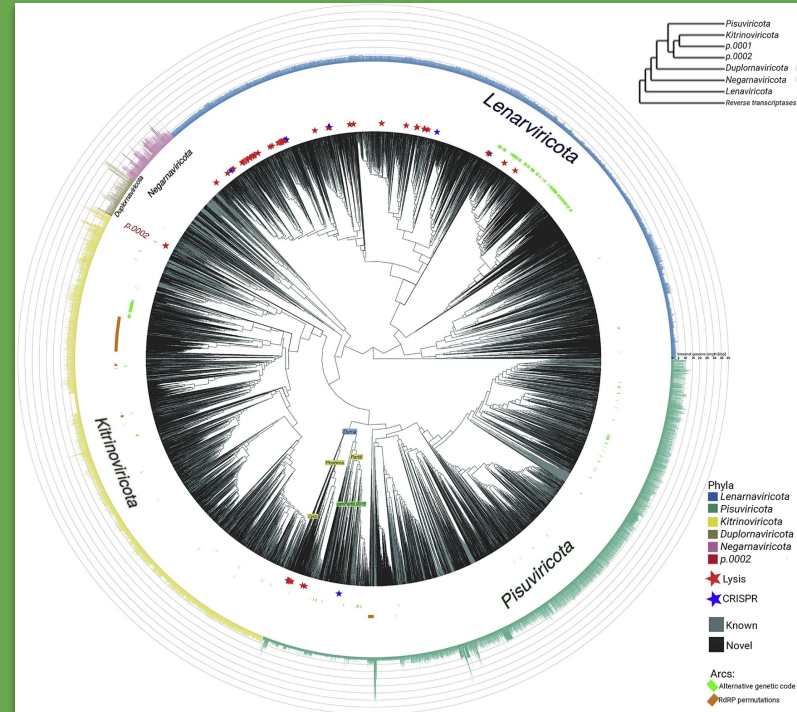
RNA Processing

- rRNA-depleted total RNA
- Sequenced and assembled; RdRP used to identify RNA viruses



Expansion of the global RNA virome reveals diverse clades of bacteriophages, Neri et. al

- Analyzed 5,150 metatranscriptomes
- Identified 2.5M+ RNA virus contigs, clustered into 124k+ viral groups
- Discovered two candidate phyla (p.0001, p.0002)
- Major expansion of RNA bacteriophages, esp. Lenarviricota
- Found CRISPR evidence of bacterial hosts



Methods

Sample Collection

Soil samples collected from heated and control plots
Separated into organic and mineral horizons

RNA Extraction and Sequencing

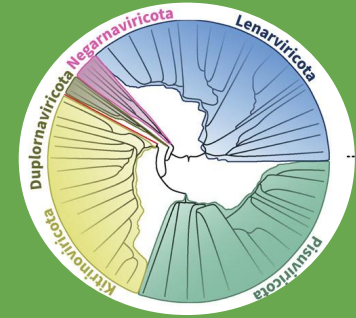
Total RNA extracted from soil
rRNA depleted and sequenced

RNA Virus Identification

Assembled contigs ≥ 1 kb
Filtered out DNA-like sequences
Identified RdRP catalytic motifs to detect RNA viruses

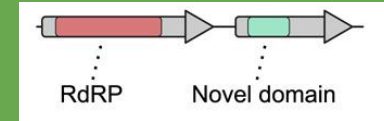
Clustering and Phylogenetic Analysis

Grouped contigs into RvANI90 clusters (90% nucleotide identity)
Assigned taxonomy using expanded RNA virus phylogeny



Community Comparisons

Viral communities compared across heated vs control
Analyzed separately for organic and mineral soils
Visualized phylum-level distributions with pie charts



RdRP Discovery

348,762 RdRP-containing RNA virus contigs identified

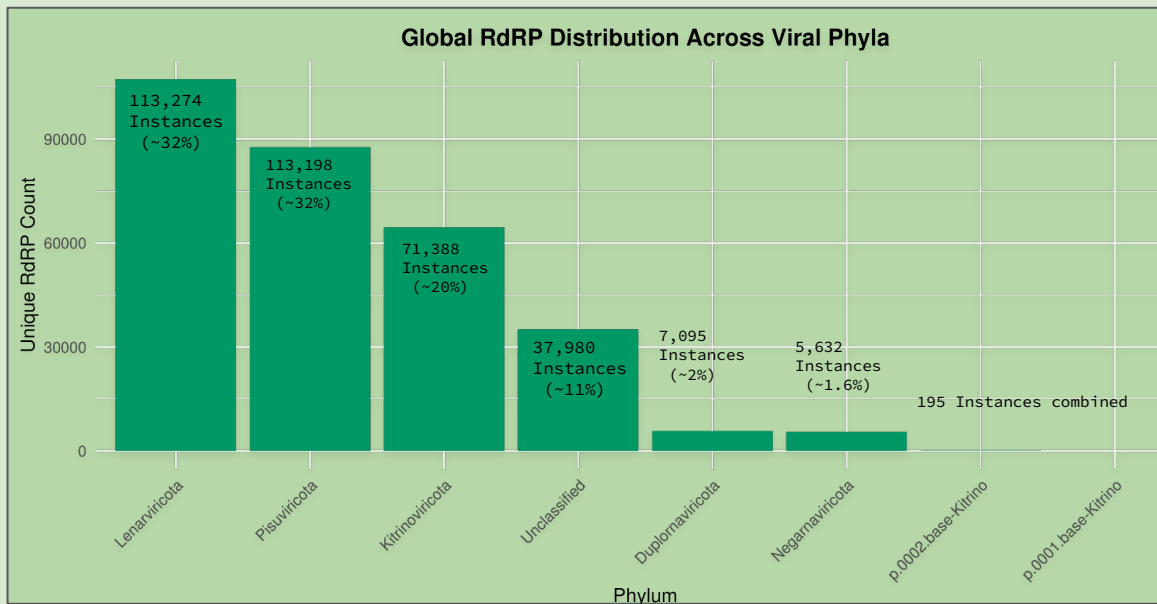
- Focused on contigs ≥ 1 kb with complete catalytic motifs

124,743 clusters formed (90% nucleotide identity)

- Majority represent novel viral lineages

Phylogenetic highlights:

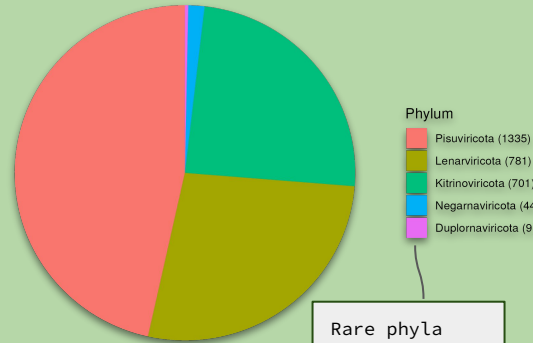
- Lenarviricota most abundant – consistent with bacterial phage dominance
- Candidate phyla (p.0002, p.0001) reveal hidden RNA virus diversity



Key Conclusions

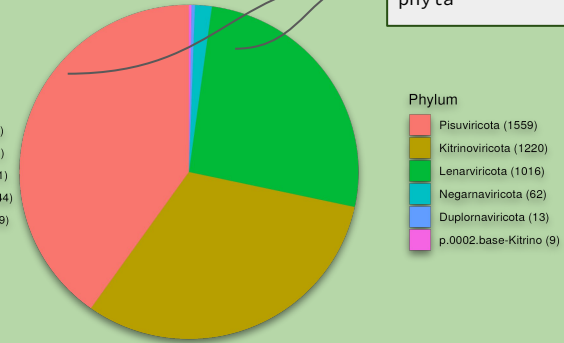
- Heating reduces overall RNA viral diversity
- Lenarviricota becomes proportionally enriched under heating
- Mineral soils dominated by major phyla
- Organic soils retain rare and candidate RNA viruses

RNA Phage Phylum Composition (Organic)
Total genomes: 2870



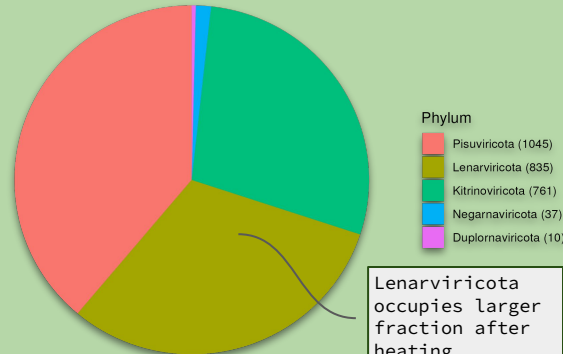
Rare phyla persist in organic soils

RNA Phage Phylum Composition (Mineral)
Total genomes: 3879



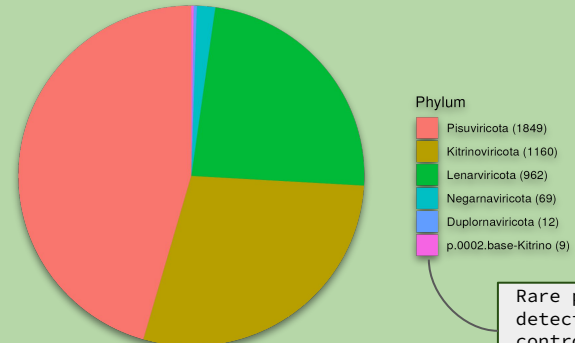
Mineral soils dominated by major phyla

RNA Phage Phylum Composition (Heated)
Total genomes: 2688



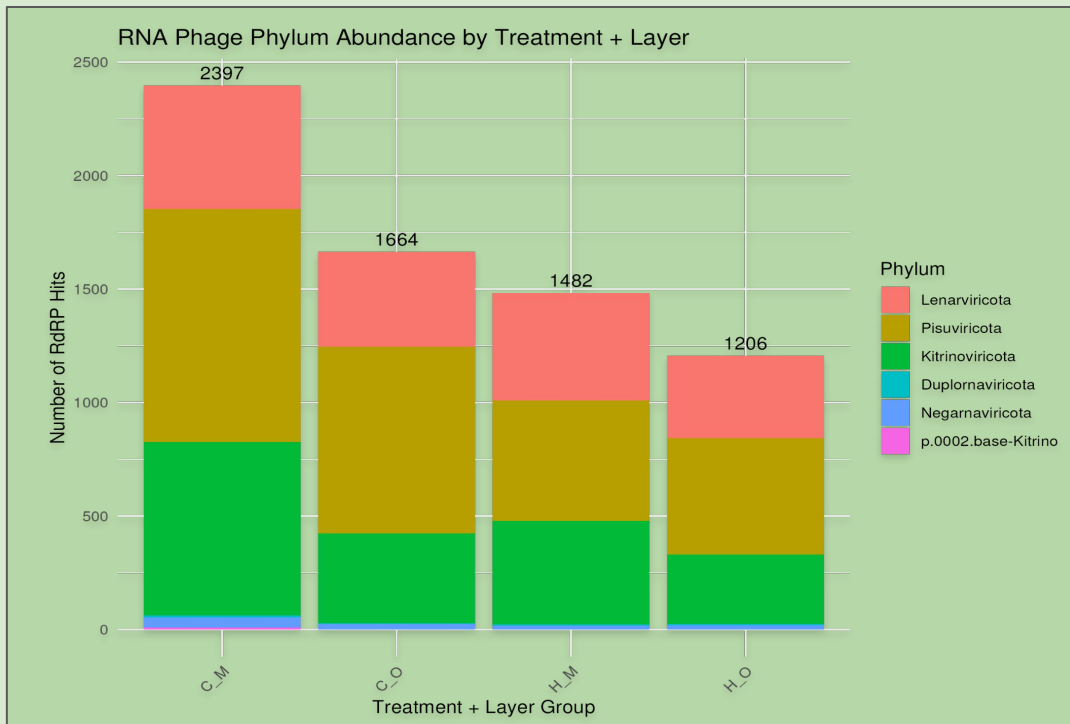
Lenarviricota occupies larger fraction after heating

RNA Phage Phylum Composition (Control)
Total genomes: 4061



Rare phyla detected in control soils

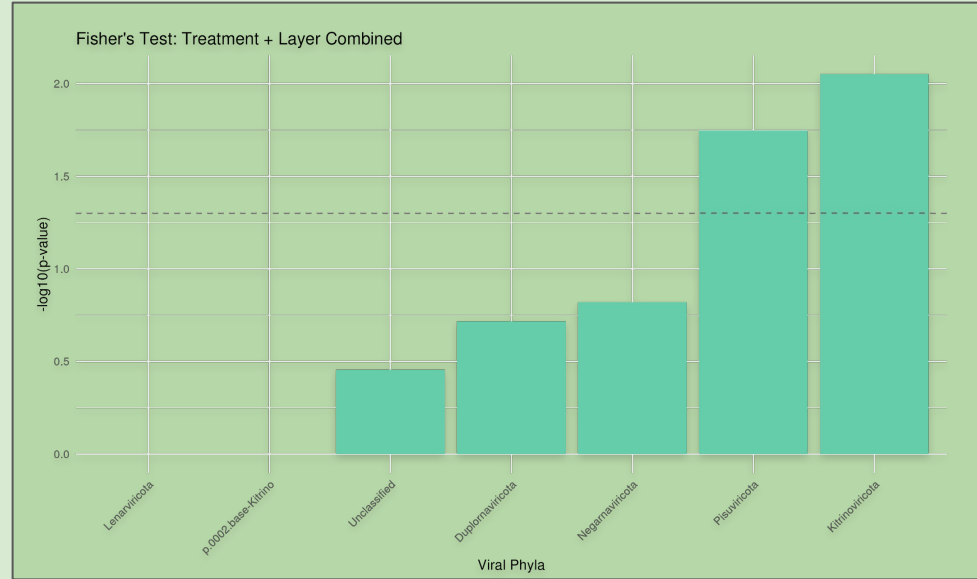
Environmental Distribution



- Control mineral soils (C.M) had the highest total viral counts and phylum richness
- Heating reduced overall RNA viral abundance across both horizons
- Organic soils (C.O and H.O) contained more rare phyla, including p.0002
- Lenarviricota and Pisuviricota remained dominant across all conditions
- Heating and depth together simplified viral community structure

Fisher's Exact Test

- Statistical significance test
- The differences between each condition is significant!
 - With exceptions of Lenarviricota and p.0002.base-Kitrino



Review on Phyla

Phylum	What We Observed	Ecological Interpretation
Lenarviricota	Abundant across all soils; enriched w/ heat	Bacterial phages thrive under warming
Pisuviricota	Stable across treatments and depths	Generalist phages with broad ecological tolerance
Kitrinoviricota	Decreased in heated soils	Loss of eukaryotic hosts (e.g., fungi, protists)
p.0002 / p.0001	Found in control, organic soils	Rare phages sensitive to heat or environmental stress

Ecological trends and viral community patterns referenced from:

Neri et al., *Cell* 2022; Callanan et al., *Sci. Adv.* 2020; Treseder et al., *Ecology Letters* 2016; Shi et al., *Nature* 2016

Review on Genes and Adaptations

Phylum	Key Genes	Adaptive Insight
Lenarviricota	Sgl proteins, M15 peptidases	Efficient bacterial lysis, thermal resilience
Pisuviricota	Capsid variants, permuted RdRPs	Broad host range, structural flexibility
Kitrinoviricota	Movement/fusion proteins	Eukaryotic targeting, heat sensitivity
p.0002	M35 peptidases, SD motifs	Bacterial niche phages, vulnerable to stress

Gene domain annotations and adaptive traits derived from:

Neri et al., *Cell* 2022; Chamakura & Young, *Curr. Opin. Microbiol.* 2020; Cahill & Young, *Adv. Virus Res.* 2019; Wolf et al., *Nat. Rev. Microbiol.* 2018

The background of the slide is a close-up, top-down view of dark brown, rich soil. The soil has a crumbly, granular texture with many small clumps and some thin, light-colored roots or organic matter visible. The lighting is even, highlighting the natural texture of the earth.

FUTURE DIRECTIONS...

Assembly Strategies

Base:
28 MetaG + 28
MetaT

Assembly

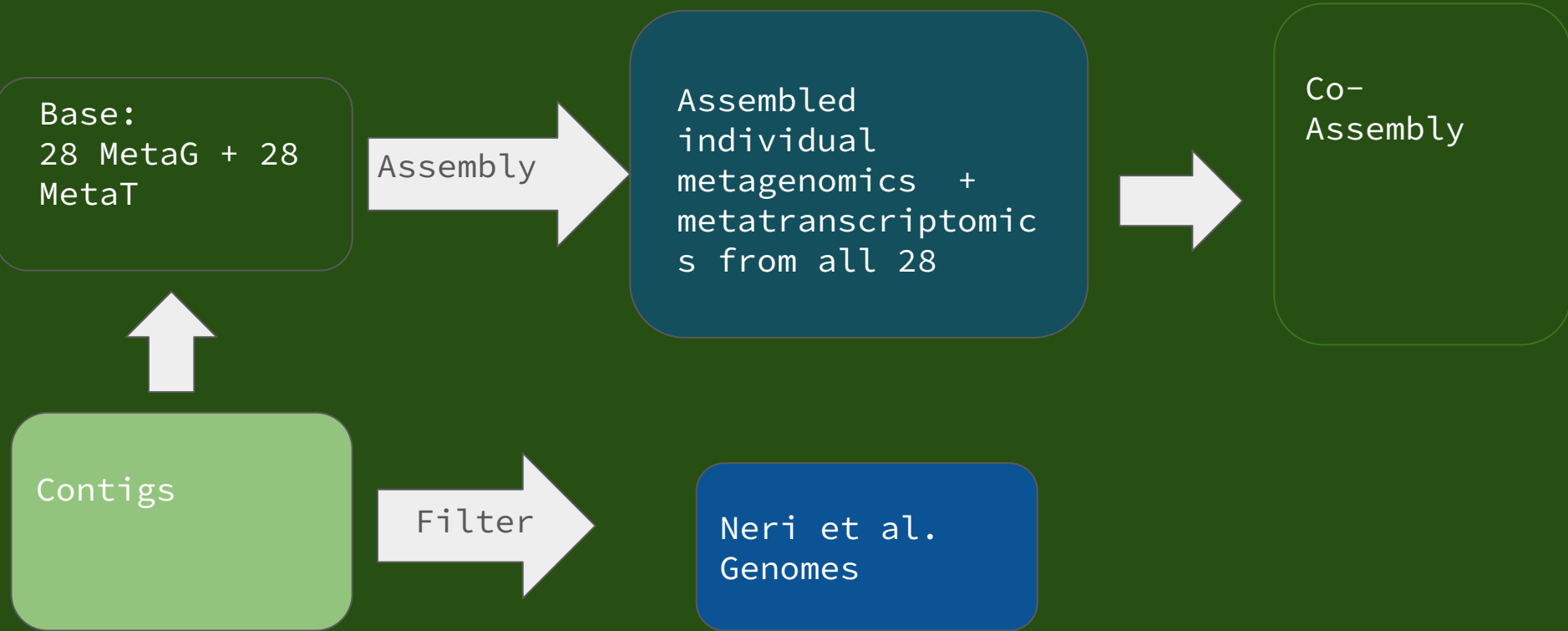
Assembled
individual
metagenomics +
metatranscriptomic
s from all 28

Co-
Assembly

Contigs

Filter

Neri et al.
Genomes



Assembly Strategies

Base:
28 MetaT

Assembly

Assembled
individual
metatranscriptomes
from all 28

Contigs

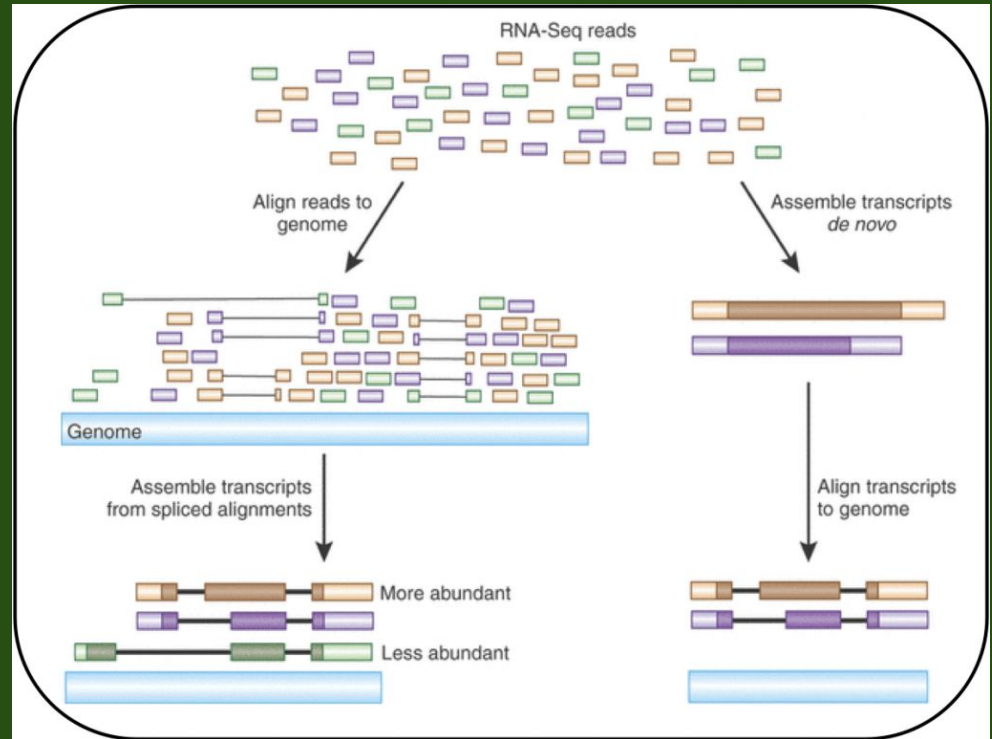
Filter

Neri et al.
Genomes



Read Mapping Steps

1. Dereplication
2. Mapping (eg. Bowtie)
3. Normalization
4. T-test
5. Visualization (eg. Tidyverse)



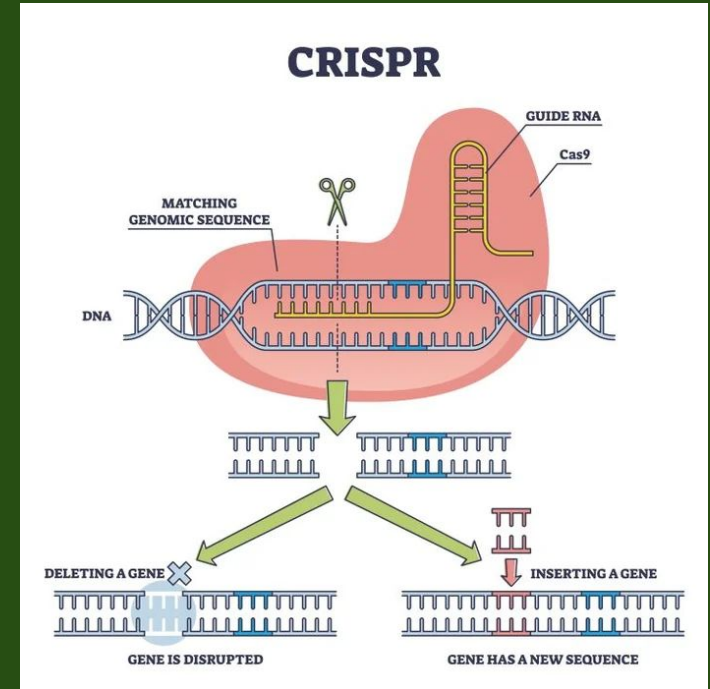
(EMBL-EBI)

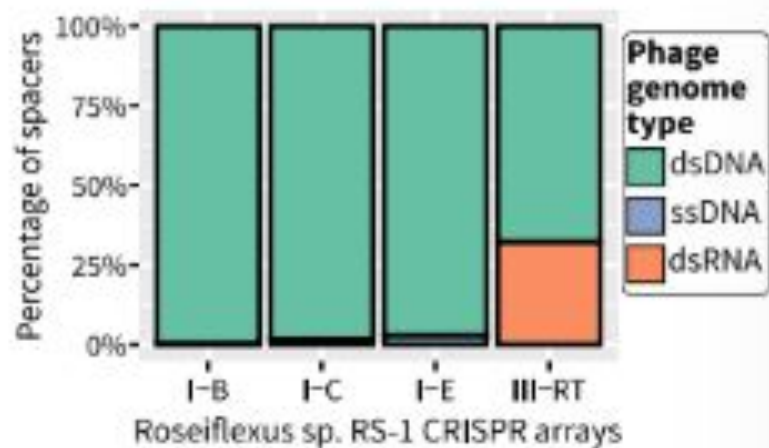
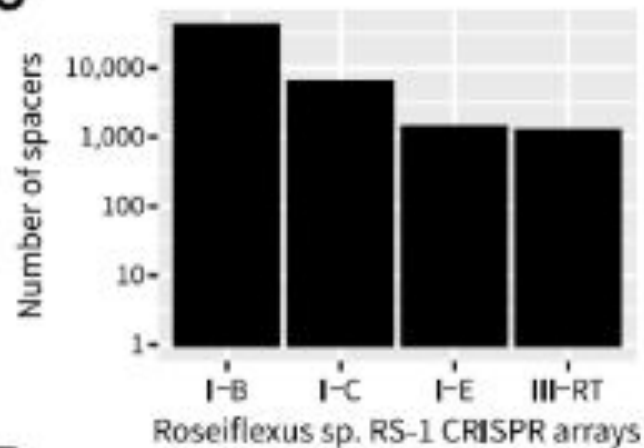
CRISPR Data

- How does CRISPR work?
- Use phages and search against bacterial CRISPR spacers
 - Find out what hosts you have!
- Find new classes of CRISPR

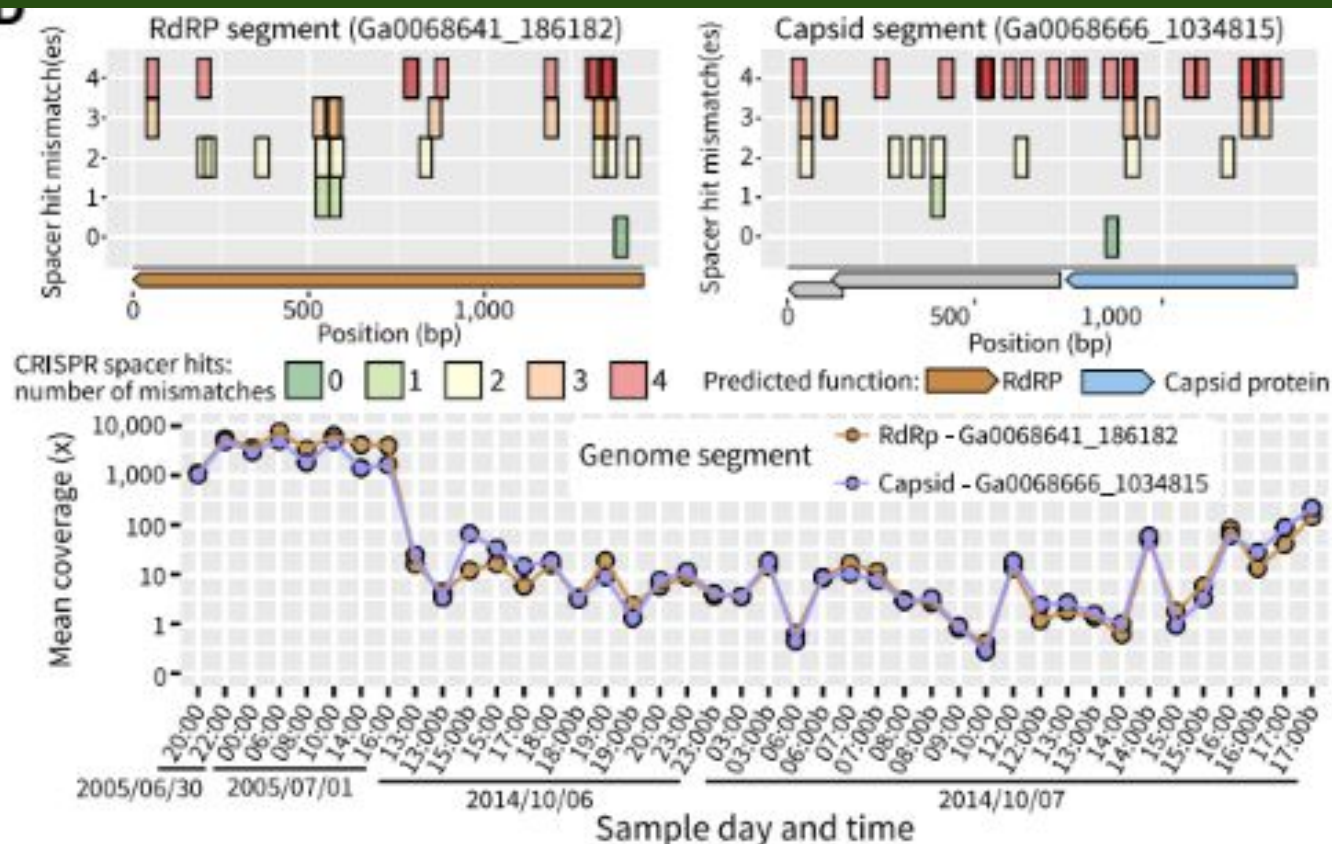
Limitations:

- Gauge Accuracy (short length)
[High substitution Rate]
- Limited to just bacteria from our samples



C

D



Limitations

- RdRP detection limited
- Non-segmented RdRPs get unreported
- More to explore in the Global virome

Thank you! Questions?